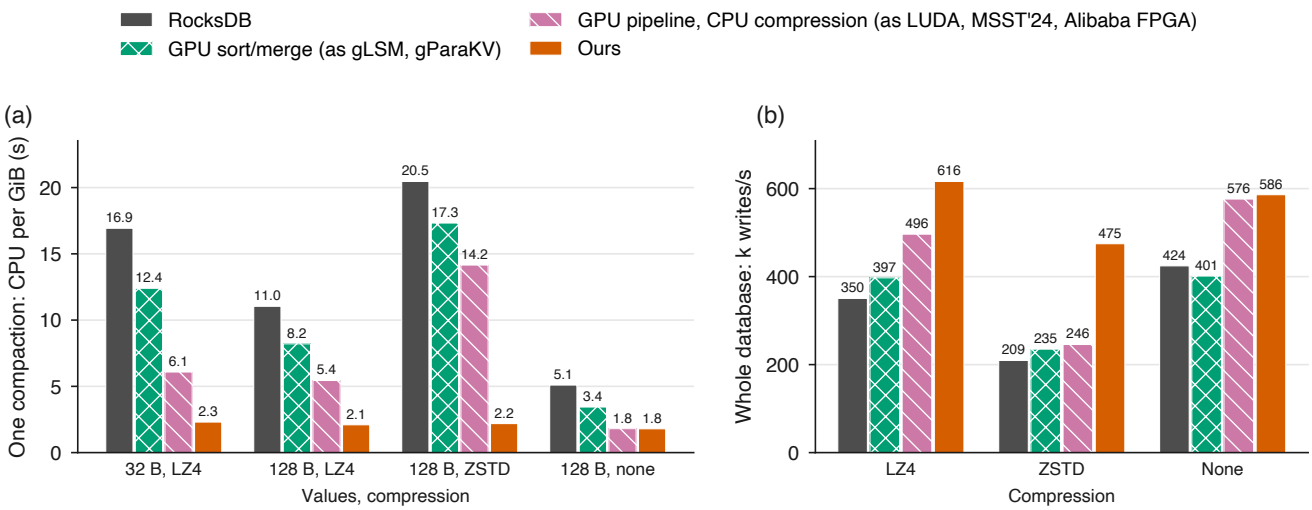
RocksDB
GPU sort/merge (as gLSM, gParaKV)
GPU pipeline, CPU compression (as LUDA, MSST'24, Alibaba FPGA)
Ours
(a)
One compaction: CPU per GiB (s)
0
5
10
15
20
16.9
12.4
6.1
2.3
11.0
8.2
5.4
2.1
20.5
17.3
14.2
2.2
5.1
3.4
1.8
1.8
32 B, LZ4
128 B, LZ4
128 B, ZSTD
128 B, none
Values, compression
(b)
Whole database: k writes/s
0
200
400
600
350
397
496
616
209
235
246
475
424
401
576
586
LZ4
ZSTD
None
Compression